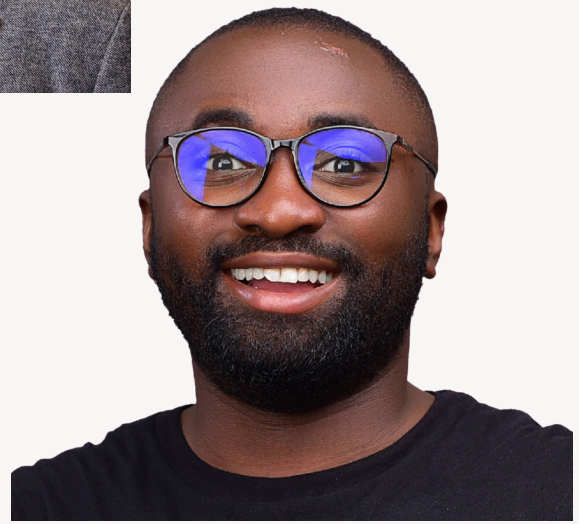
A DBA's Surprising Journey To Databases On Kubernetes

Shivadeep Gundoju (ING)

Divine Odazie (EverythingDevOps)





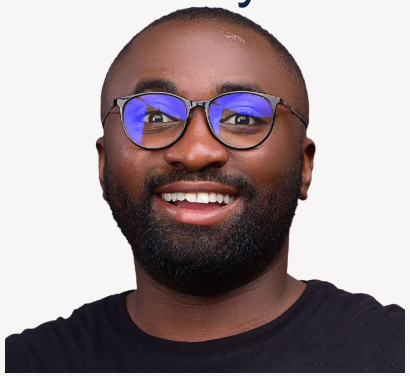
Shivadeep Gundoju

- DBaaS Engineer at ING
- Oracle Certified Professional
- IaC, Data on Kubernetes enthusiast



Divine Odazie

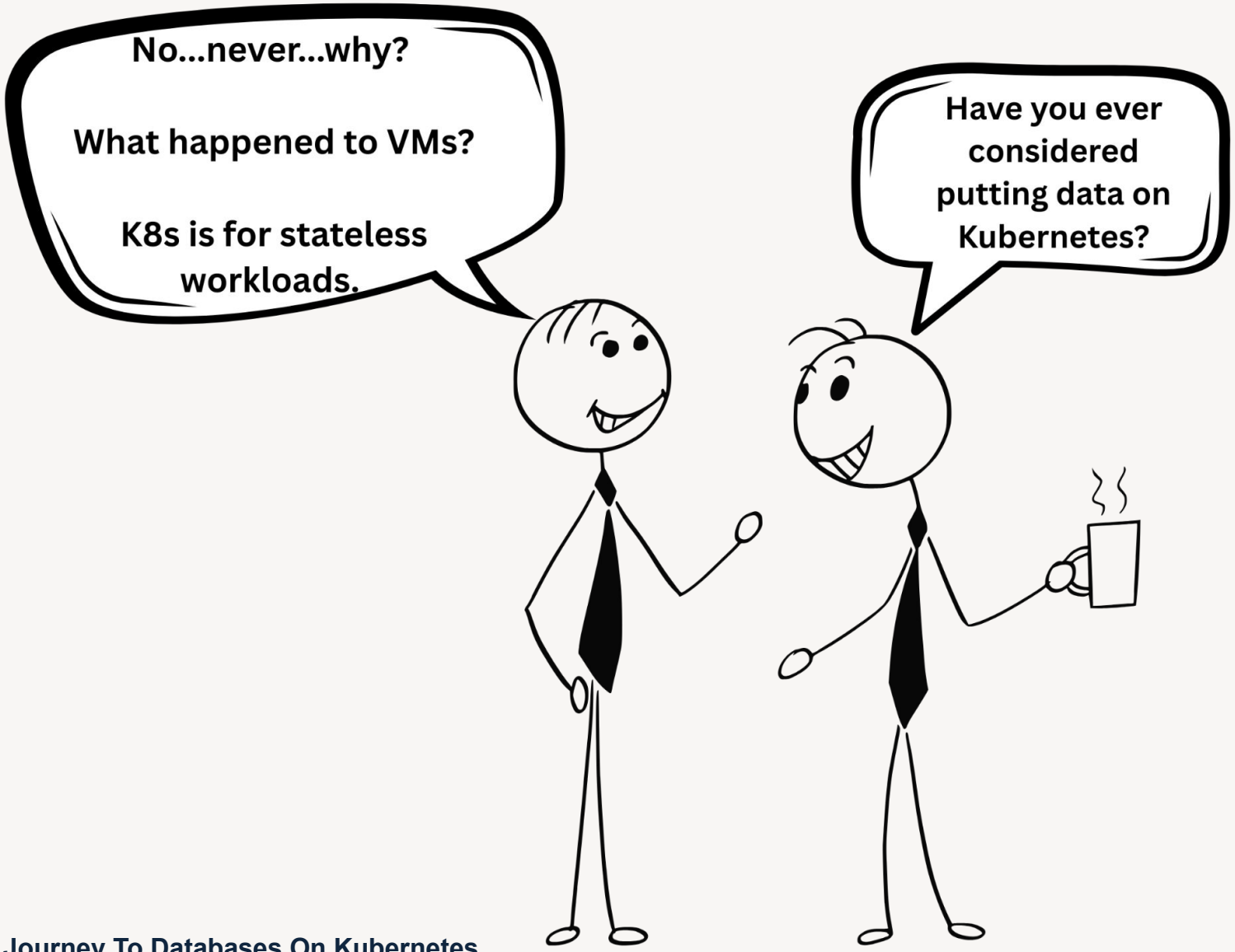
- Founder & CEO EverythingDevOps
- Data on Kubernetes Ambassador
- Electrical/Electronics Engineer
- Certified Kubernetes Engineer (Almost Kubestronaut)
- I’m excited to speak at SREday again!!



Raise your hand if you support the idea of databases on Kubernetes

Raise your hand if you have ever deployed a databases on Kubernetes

A Conversation I Had At SREday London 2024



ING at a glance

Our purpose

Empowering people to stay a step ahead in life and in business.

Our priorities

Sustainability at the heart

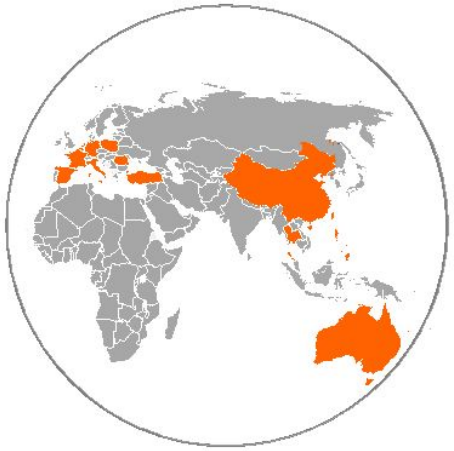
Superior customer experience

We serve 37 million customers in more than 40 countries

Our **Market Leaders**



Our **Challengers & Growth markets**



Market Leaders

- Netherlands*
- Belgium
- Luxembourg

Challengers Markets

- Australia
- France **
- Germany
- Italy
- Spain

Growth Markets

- Poland
- Romania
- Turkey
- Philippines **
- Stakes in Asia

(*) ING's corporate head office is located in Amsterdam, The Netherlands

(**) In 2022, ING discontinued its retail activities in these markets

Wholesale Banking
international network and global franchises

Map highlights countries where ING has an office

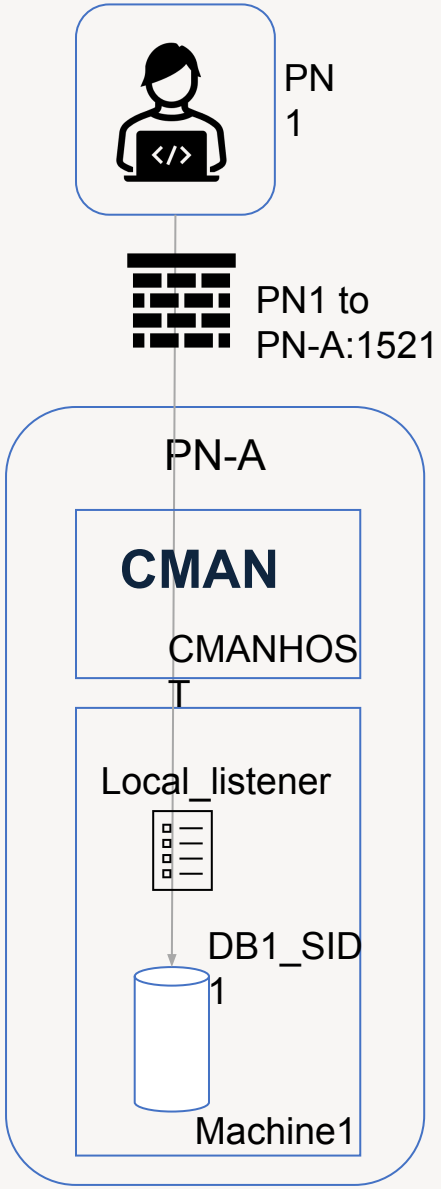




Coming up..

1. What is CMAN?
2. CMAN on Container
3. CMAN on Kubernetes ?
4. Demo
5. Oracle Database on Container&Kubernetes ?

Oracle Connection Manager(CMAN)



- Access Control
- Connection Filtering and logging
- Transparent to DB migrations
- Port & Protocol Translation
- Traffic Routing & Load Balancing

Connection=(**Host:CMANHOST** Port:1521
SID:DB1_SID1)

Connection=(Host:Machine1 Port:1521
SID:DB1_SID1)

CMAN on Containers?:

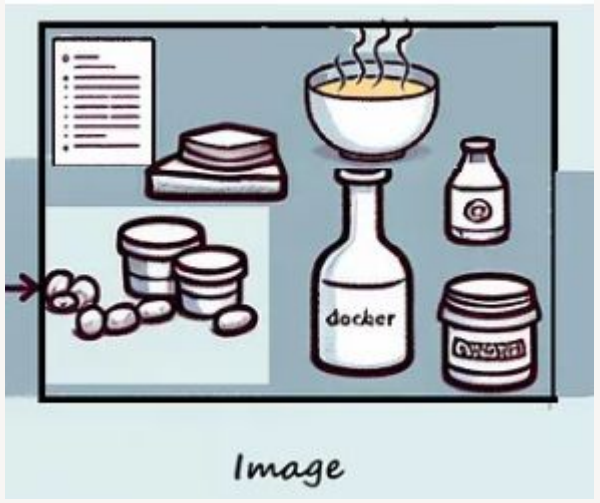
Is it a candidate to be in a container?

Is it even possible ? And How?

what ?



BUILD

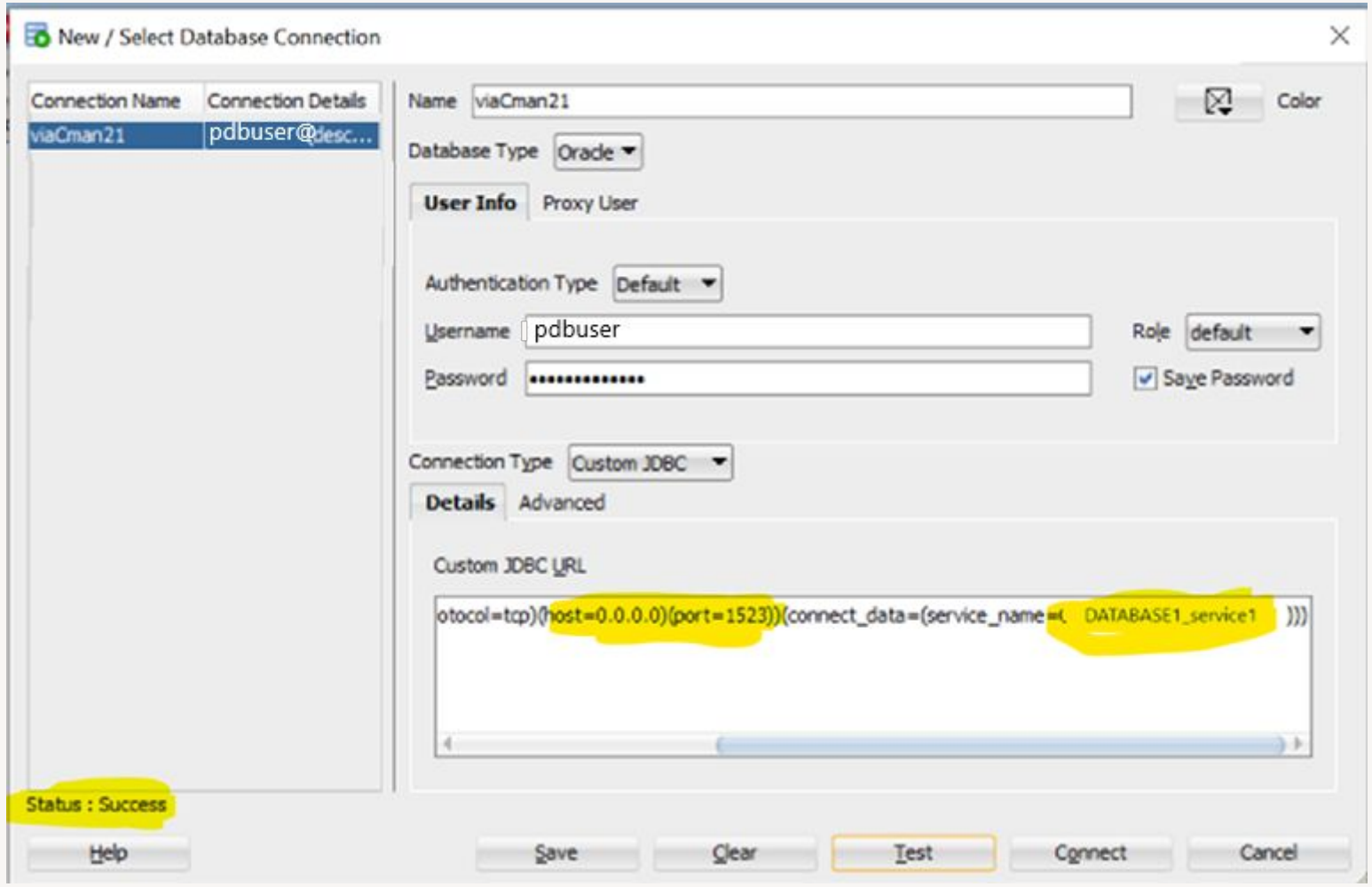


RUN



Source: https://container-registry.oracle.com/ords/ocr/ba/database/cman

CMAN on Local Machine



Source: https://www.oracle.com/database/kubernetes-for-container-database/

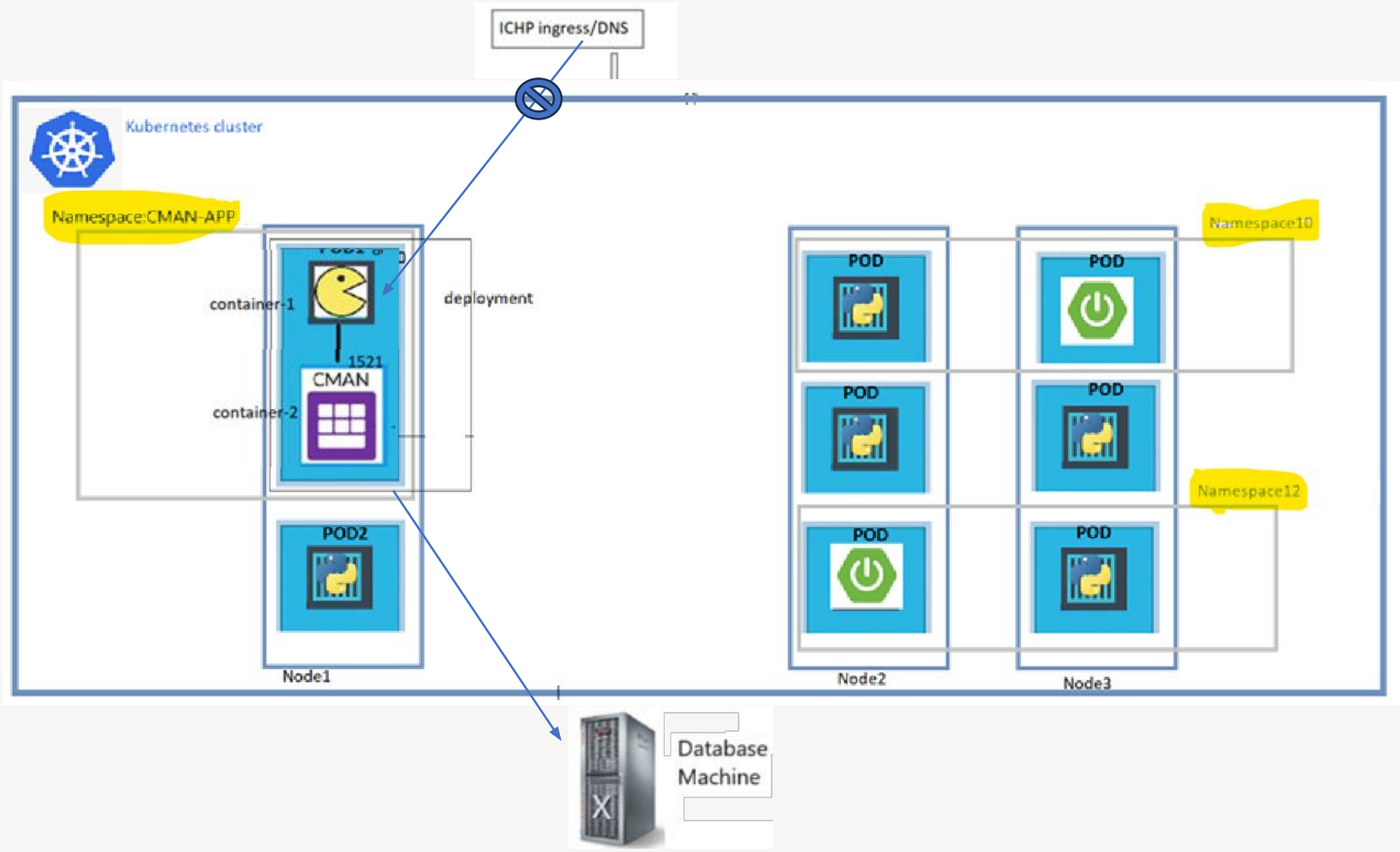
CMAN on Kubernetes

- Why?
- What?
- How?
  - ICHP-ING Container Hosting Platform



Source: https://github.com/ing-bank/quota-scaler

Why PACMAN ?


ICHP ingress/DNS
Kubernetes cluster
Namespace:CMAN-APP
container-1
deployment
1521
CMAN
container-2
POD2
Node1
POD
POD
POD
POD
POD
POD
Namespace10
Namespace12
Node2
Node3
Database
Machine

Databasemachine

```
bash-4.4$
```

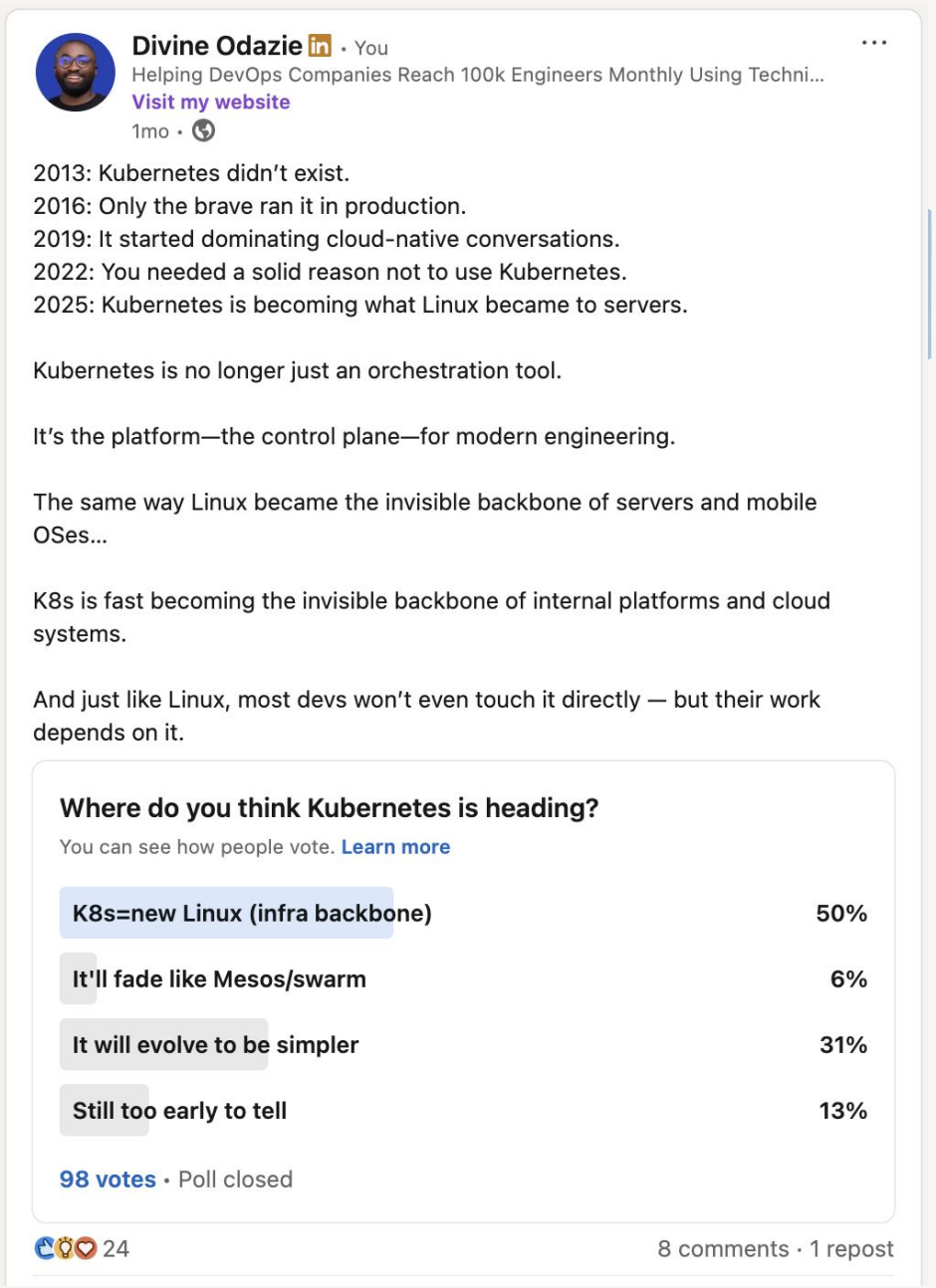
Oracle Database on Container/Kubernetes:

- Yes, Starting 19c
- OraOperator : Create/Manage oracle database with native Kubernetes

Source: https://www.oracle.com/database/kubernetes-for-container-database/



How DBAs can evolve to manage databases on Kubernetes.



In summary

- From Skepticism to Adoption
- As always learning by doing
- Evolving the DBA Role
- The Future…Kubernetes the new **Linux**?

Thank you

Questions?







